DATA SCIENCE & AI · MILESTONE 2

Talk to Your Database

Dataset Preparation, Profiling & Exploratory Data Analysis
on the Spider Text-to-SQL Benchmark

TEAM MEMBERS

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INTRODUCTION

The Spider Text-to-SQL dataset contains many SQLite databases spanning diverse real-world domains, each paired with natural language questions and their SQL queries. Before applying any modeling, we first need to understand its schemas, attribute distributions and data-quality characteristics through exploratory data analysis - the focus of this milestone.

OBJECTIVES OF MILESTONE 2

1. Identify and acquire the required datasets
2. Understand database schemas and data characteristics
3. Perform exploratory data analysis (EDA)
4. Assess data quality; identify missing / inconsistent values
5. Generate metadata and profiling reports
6. Produce a training-ready dataset with documented preprocessing steps

DATASET IDENTIFICATION

Spider: Cross-Domain Text-to-SQL Benchmark

A large-scale, human-labeled dataset for complex and cross-domain semantic parsing and Text-to-SQL generation, purpose-built for training and evaluating NL2SQL systems across many application domains.

SOURCE

yale-lily.github.io/spider · github.com/taoyds/spider

LICENSE STATUS

Publicly available for research purposes under its published licensing terms

PURPOSE

Training and evaluating Text-to-SQL systems across multiple domains

ALTERNATIVES CONSIDERED

WikiSQL and CoSQL were evaluated and set aside in favor of Spider

KEY CHARACTERISTICS

- Multiple real-world relational database domains
- Complex SQL query structures
- Cross-domain database schemas
- Human-generated natural language questions
- Supports evaluation of Text-to-SQL model generalization

DATASET DESCRIPTION

Scale, Features & Formats

~200	130+	1000s	10000+
SQLite databases	Database domains	Tables & columns	NL questions + SQL queries

FEATURES

- Database tables
- Column names
- Data types
- Foreign key relationships
- Primary keys
- SQL queries
- Natural language questions

TARGET VARIABLE

SQL query corresponding to a given natural language question

DATA FORMATS

- SQLite (.sqlite)
- JSON (.json)
- CSV (generated during preprocessing)

SCHEMA

Each database contains multiple relational tables connected through primary and foreign keys.

Includes:

• JSON schema files • Database summary CSV

• Column profiling CSV

Metadata captured:

• No. of tables & columns • Data types • Primary & foreign keys and relationships • Attribute characteristics

Licensing, Privacy, Quality & Ethics

1

Data Source & Licensing

Obtained from the official Spider project repository; publicly available for academic and research use.

2

Privacy

Contains no personally identifiable information. Databases are synthetic or anonymized, intended for research.

3

Data Quality

Validated via SQLite integrity checks, tables and column profiling, missing-value identification, distinct-value counts, and encoding checks.

4

Ethics & Bias

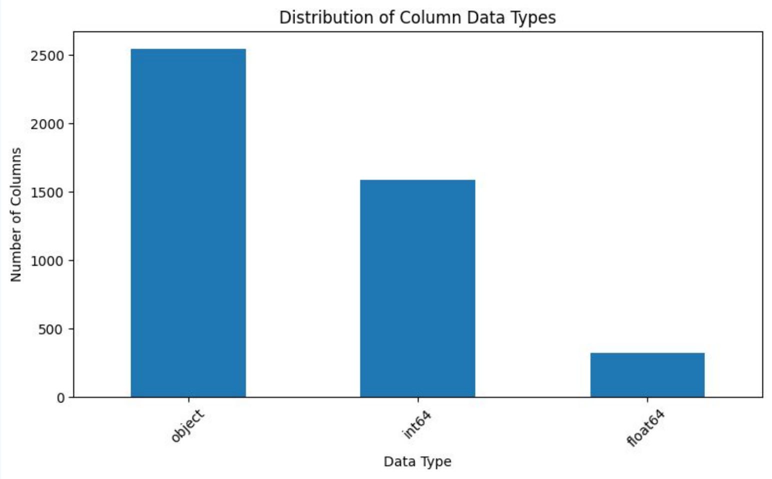
Uneven domain representation, SQL complexity variation, and limited industry coverage; dataset intended solely for research and education.

All preprocessing scripts were documented and version-controlled. The generated datasets can be reproduced by executing the provided Python scripts on the original Spider dataset.

Data Types & Missing Values

Data Type Distribution Analysis

Distribution of attribute data types across database columns.



Missing Value Analysis

Columns with missing entries, counts, and % of incomplete records.

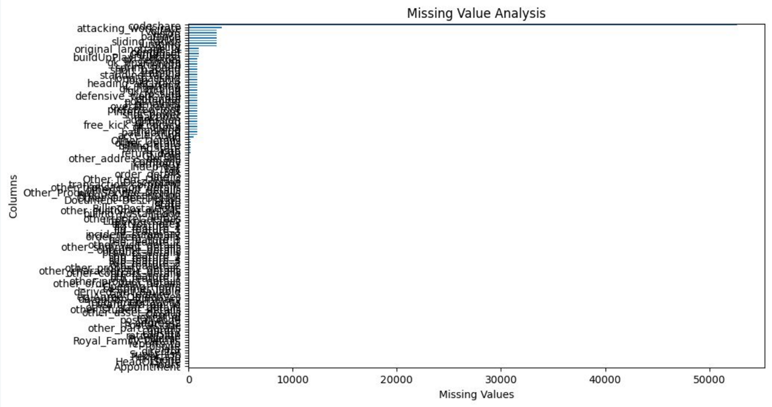
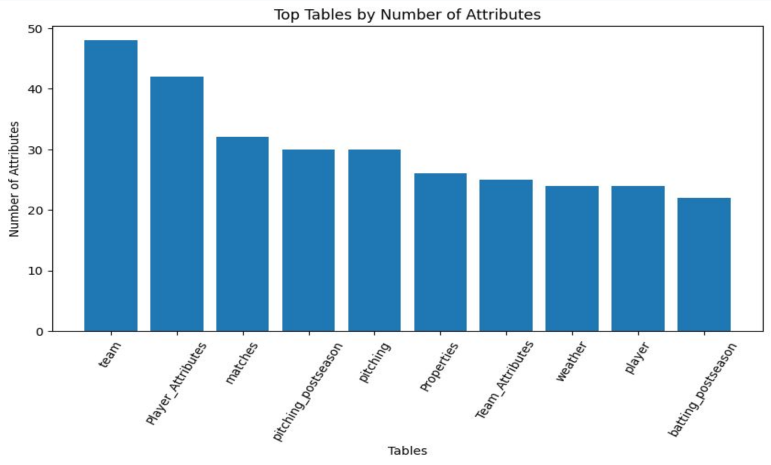


Table Complexity & Attribute Cardinality

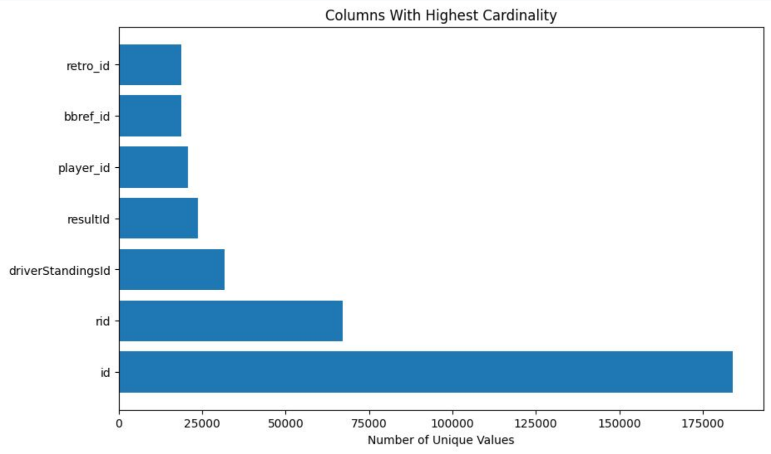
Table Complexity

Number of attributes per table - more columns indicate higher schema complexity.



Attribute Cardinality

Distinct-value analysis highlights high-uniqueness columns - primary and transaction identifiers.



Preprocessing, Integration & Splitting

PREPROCESSING STEPS COMPLETED

- Automated database discovery
- Schema extraction
- Table profiling
- Missing value detection
- Data type identification
- Removal of unreadable records (invalid encodings)
- Standardized export to CSV format
- Metadata generation

Since this milestone focuses on database profiling rather than predictive modeling, normalization, feature scaling and categorical encoding were not required.

DATASET INTEGRATION

- Multiple SQLite databases were processed independently.
- No direct merging was performed - each database represents an independent application domain with its own schema.
- Schema consistency was maintained via standardized metadata.

DATASET SPLITTING

The Spider dataset already provides predefined training and development splits. The official partitions were retained to ensure consistency with published benchmark evaluations.

Outputs & Deliverables

The final prepared dataset consists of:

- Original SQLite databases
- Extracted schema metadata
- Database summary reports
- Column profiling reports
- Cleaned metadata exports

The dataset is now organized and documented, enabling **direct use** in future machine learning experiments.

The following deliverables were generated:

- Database summary report (*database_summary.csv*)
- Column profiling report *(column_summary.csv)*
- Schema metadata *(schema.json)*
- Automated preprocessing scripts
- Documentation of preprocessing workflow
- Training-ready database collection

Challenges & Key Observations

CHALLENGES ENCOUNTERED

- Managing hundreds of SQLite databases
- Handling varying database schemas
- Processing inconsistent text encodings
- Automating metadata extraction across diverse domains
- Generating consistent profiling outputs

KEY OBSERVATIONS

- The dataset is composed of many applications, each with its own schema
- Table sizes and column counts differ greatly across databases
- Most databases show excellent structural consistency, aside from a few with text-encoding issues

LOOKING AHEAD

Planned Activities for Milestone 3

01

Feature Engineering

Prepare engineered features tailored for Text-to-SQL tasks

02

Baseline Modeling

Train baseline Text-to-SQL models on the prepared dataset

03

Model Evaluation

Evaluate models using official Spider benchmarks

04

Error Analysis

Analyze errors and optimize model performance